Superimposition methods: what the implants taught us

Literature review topic 1 of 19: Superimposition Methods · Nielsen 1989, AJODO 95:422–431: maxilla, 2D, n = 18 · Doppel, Damon, Joondeph, Little 1994, AJODO 105:161–168: maxilla, 2D, n = 50 · Nguyen et al. 2018, AJODO 153:747–754: mandible, 3D, n = 20 + 25 · Presenter, date, department

Why superimposition matters

- The question: did the molar erupt, or did the maxilla descend around it?
- Overall superimposition = total change; regional = movement within the jaw; difference = growth.
- Register on a structure that moved and its movement is subtracted from everything.
- Palatal best fit hid 1.33–1.84 mm of 4-year eruption (Nielsen, Table IV).
- Tracing error 0.5 mm; head rotation shifts a zygomatic implant 3 mm (Doppel, Methods).

The gold standard: Björk's implants

- Tantalum pins in bone, serial films: Björk 1955 (reference 1 in Doppel and Nguyen).
- Maxillary sites: below ANS, zygomatic process, palate–alveolar border medial to U6 (Doppel, Methods).
- Nasal floor resorbs, orbital floor apposes, ratio 3 : 2 (Björk & Skieller, via Nielsen p. 423).
- Anterior zygomatic surface stable in all 9 cases (Björk & Skieller 1977, via Doppel p. 162).
- Implants no longer placed (ALARA, IRB; Nguyen): bone plates used as regional implants.

Nielsen 1989: aim and method

- Aim: how far do two anatomic methods mis-read 4 years of growth versus implants?
- 18 of 35 UCSF implant subjects (6 boys, 12 girls), selected on film and implant visibility.
- T1 10 ± 0.37 y; interval 4 ± 0.56 y; treated and untreated mixed, mostly Class II.
- Three methods: best fit (palatal plane at ANS), implant, structural (anterior zygomatic surface).
- Structural method held apposition equal to resorption, 1 : 1 (Methods); Downs grid; second observer.

Nielsen 1989: results and take-home

Landmark	Implant	Best fit	Implant – best fit	Implant – structural
PNS	−0.70 ± 1.14	+0.36 ± 0.53	1.08, P < .001	−0.33, NS
ANS	−1.62 ± 0.83	+0.28 ± 0.70	1.88, P < .001	−0.17, NS
A point	−2.33 ± 1.17	−0.47 ± 1.16	2.24, P < .001	−0.04, NS
U6 cusp	−4.75 ± 2.13	−3.40 ± 1.68	1.33, P < .001 or better	−0.29, NS
UI edge	−3.39 ± 2.72	−1.40 ± 2.30	1.84, P < .001 or better	−0.20, NS

- vertical change over 4 years, mm, negative = downward (Tables I–IV)
- Horizontal: implant − structural 0.53–0.66 mm, P < .01; A point 0.41 mm, NS (Tables III–IV).

Doppel 1994: aim and method

- Aim: which maxillary structures stay closest to a three-implant superimposition over ≥ 3 years?
- 50 of ≈ 700 UW implant records (≈ 7 %, derived); 23 M, 27 F; films 1967–1975.
- T1 11.9 ± 1.4 y, T2 16.0 ± 1.7 y, interval 4.2 ± 0.96 y (Table I); majority treated.
- Metric: maximum deviation of 17 structures, calipers to 0.01 mm; tracing error 0.5 mm.
- Figs 1–6: one case traced by six anatomic methods with the implants drawn in.

Doppel 1994: results and take-home

Structure	n	Mean ± SD	Max	Note
Internal palate	50	0.85 ± 0.43	2.45	best number, poor empirical fit
Anterior zygomatic contour	40	0.95 ± 0.57	2.56	traceable 80 %; direction variable
Posterior zygomatic contour	50	0.97 ± 0.43	3.12	traceable 100 %; always posterior
Palate, superior / inferior border	50	1.05 ± 0.52 / 1.14 ± 0.60	2.90 / 3.34	listed not reliable
ANS, vertical / horizontal	50	1.29 ± 0.86 / 1.43 ± 1.14	4.82 / 5.09	
Orbital floor	47	1.40 ± 0.98	4.39	
Infraorbital foramen	34	1.67 ± 1.31	5.79	text says traced in 32 %
A point	50	2.13 ± 1.63	6.55	
PNS	27	2.27 ± 1.75	9.10	visible in 52 %
PTM, vertical / horizontal	50	2.39 ± 1.69 / 1.52 ± 0.99	8.81 / 4.10	
Cranial base rows (3)	50	2.52–2.64	up to 11.99	maxilla rotates vs base; lever-arm error (p. 167)

Nguyen 2018: aim and method

- Aim 1: which Björk mandibular regions stay stable in 3D against bone plates?
- Aim 1: 20 Class III bone-anchor patients, 12.1 ± 1.3 → 13.3 ± 1.1 y; plates at incisor apices.
- Regions: chin (pogonion to B point), symphysis inner cortex, third-molar crypt; canal excluded a priori.
- Voxel registration (Slicer CMF, ITK-SNAP); i-CAT or NewTom 3G; 0.3 mm voxels; absolute mean surface error.
- Aim 2: 25 patients, 12.7 ± 1.4 y, CVM 2–5, 12.6 ± 0.9 months; two observers; ICC.

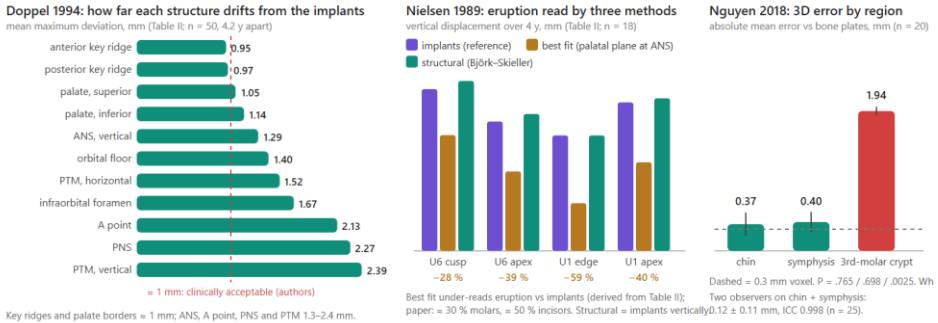
Nguyen 2018: results and take-home

Region	Error, mm	P
Chin (pogonion to B point)	0.37 ± 0.16	.765
Symphysis, inner cortex	0.40 ± 0.14 (abstract ± 0.15)	.698
Third-molar crypt	1.94 ± 0.06	.0025

- absolute mean surface error against the bone-plate registration (Table; Results)
- Interobserver error, chin + symphysis: 0.12 ± 0.11 mm (abstract misprints ± 1.1); voxel 0.3 mm.

What the three agree on

- Implants or plates are the truth; every anatomic method is a proxy with measurable bias.
- ANS, PNS, PTM, A point never registration points (Doppel 1.3–2.4 mm; Nielsen under-read 1.1–2.2 mm).
- Anterior zygomatic surface is the maxillary reference (Nielsen's method; Doppel 0.95 ± 0.57 mm).
- Same ratio: Björk & Skieller 3 : 2, Doppel 1.5 : 1, ABO 3/5 : 2/5 (literature, UNVERIFIED).
- Scatter comes from locating landmarks (Nielsen, Discussion); Björk's chin and symphysis survive 3D (Nguyen).



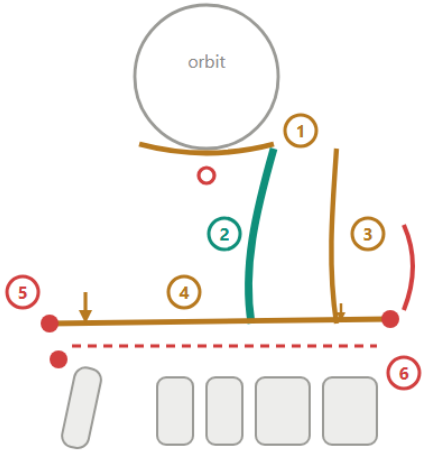
Where they differ

Issue	Nielsen 1989	Doppel 1994	Nguyen 2018
Anterior zygomatic surface	No resorptive remodelling seen (n = 18, Discussion)	Direction variable: 27.5 % anterior, 35 % posterior, 37.5 % none	—
Vertical ratio	1 : 1 executed; 3 : 2 cited (Fig 2)	1.5 : 1 recommended; cites Nielsen as agreeing	—
Posterior zygomatic contour	Reject: appositional (Riedel's method)	Keep as "most reliable" though posterior drift in 50 of 50	—
Palatal registration	Best fit at ANS hides 30–50 % of eruption	Internal palate 0.85 mm, best number, "limited value"	—
Third-molar crypt, canal	—	—	Crypt fails (1.94 mm); canal excluded; Björk used both (cited)
Verdict	Structural method valid and reliable	No maxillary area completely reliable	Chin + symphysis stable and reliable

Stable and unstable structures: the map

Stable: register here Unreliable: do not register here Remodels with growth (apposition or resorption)

Maxilla (lateral view, anterior to the left)



1 Orbital floor
apposition with growth

2 Anterior zygomatic surface
key ridge: STABLE (Björk)
no resorption (Nielsen)
drift 0.95 mm (Doppel)

3 Posterior zygomatic surface
appositional; drift 0.97 mm

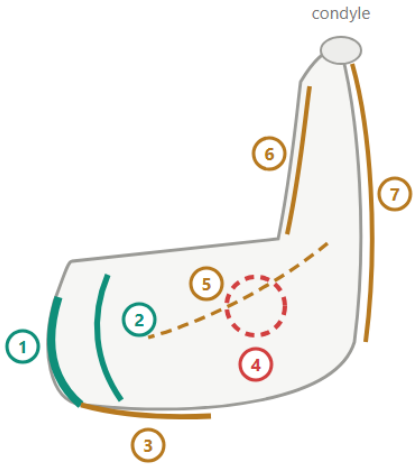
4 Nasal floor / palatal plane
resorbs; ANS ≈ 2× PNS
3:2 (Björk); 1.5:1 (Doppel)

5 ANS, A point, PNS, PTM,
infraorbital foramen: unreliable
drift 1.3–2.4 mm

6 Palate / best fit at ANS
eruption under-read 30–50 %
(Nielsen 1989)

Mean maximum deviation from implant superimposition (Doppel 1994, Table II, n = 50).

Mandible (lateral view, anterior to the left)



1 Chin anterior contour
STABLE (Björk)
3D error 0.37 ± 0.16 mm

2 Inner cortex of symphysis
STABLE; 3D error 0.40 ± 0.14 mm

3 Inferior border
deposition ≈ 2 mm near menton

4 Third-molar crypt
3D error 1.94 ± 0.06 mm
fails 16 of 18 (Ruellas 2016)

5 Mandibular canal
moves with growth; excluded

6 Anterior ramus: resorption

7 Posterior ramus: deposition

Observers on 1+2: 0.12 mm,
ICC 0.998 (Nguyen, n = 25)

Absolute mean error vs bone plates (Nguyen 2018, n = 20, 1 y). Register on 1 + 2; avoid 4 and 5.

- Maxilla, stable: anterior zygomatic surface (AP); posterior contour; orbital : nasal floor 3 : 2 vertically.
- Maxilla, unstable: ANS, A point, PNS, PTM, foramen, palate borders; palatal plane rotates +0.92°, SD 2.1°.

Reading our own cephalograms and CBCTs

- First question for any change value: registered on what?
- Palatal registration under-reads upper eruption: ≈ 0.3–0.6 mm/y at 10–14 y (derived, Nielsen Table III).
- 2D resolves 0.5 mm (Doppel): group differences below 0.5 mm are not readable on tracings.
- Mandible: chin + inner symphysis; corpus axis errs ≥ 4 mm in ≈ 10 % (Springate & Jones, registry).
- Khaleel 2026 (registry): palatal vault + corpus axis over 44–94 d; arm differences 0.2–0.35 mm, sub-resolution.

Limitations across the topic

- All retrospective, best-film samples: 50 of ≈ 700; 18 of 35; database scans minus artefacts.
- Treated samples: Doppel majority, Nguyen all; Nielsen mixed and could not separate them.
- Pubertal intervals only (4.2 y, 4 y, ≈ 1 y): no adults, no 3D data beyond one year.
- Gold standards carry error: closely spaced implants, not all bilateral, loaded plates checked clinically only.
- Reporting slips: Nielsen Table IV footnote; Nguyen ± 1.1 vs ± 0.11; Doppel 32 % vs 34/50.

What the textbook says

- Proffit Ch 6: cranial base; maxilla "maximum contour" (Fig 6.59); mandible on symphysis (Fig 6.60) (study pages, UNVERIFIED).
- Proffit Ch 4: implants show internal rotation of both jaws, masked by surface remodelling (study pages, UNVERIFIED).
- Proffit Ch 2: implant radiography "largely superseded by 3-D imaging" (study pages, UNVERIFIED).
- ABO maxillary method: zygomatic processes, orbital and palatal floors, 3/5 : 2/5 (literature, UNVERIFIED) = Doppel.
- ABO mandibular list keeps canal and third-molar bud (literature, UNVERIFIED); Nguyen drops both in 3D.

Open questions and the future

- No 3D maxillary region validated against implants; current work tests reproducibility only (literature, UNVERIFIED).
- Graf 2022 review: 27 studies, 25 at high risk of bias; no method proved accurate (literature, UNVERIFIED).
- Chen 2020: chin and symphysis < 1.0 mm over 4.6 y; foramina 3.0–3.4 mm (literature, UNVERIFIED).
- Automated registration: AI RMSE 0.48 ± 0.35 vs observer 0.38 ± 0.22 mm, NS (Vasileiou 2026, literature, UNVERIFIED).
- Dose: growth monitoring is not a Proffit CBCT indication; ≈ 0.1 vs 0.005 mSv (study pages, UNVERIFIED).

Summary

- Palatal best fit hides ≈ 30 % of molar, ≈ 50 % of incisor eruption over 4 years (Nielsen).
- Maxillary reference: zygomatic process; orbital floor rises 1.5× what the palate lowers (Doppel).
- ANS, PNS, PTM, A point and the palatal plane are not registration points (both).
- 3D mandibular reference: chin + inner symphysis, 0.37–0.40 mm over ≈ 1 y; 0.12 mm interobserver (Nguyen).
- 2D tracing resolves 0.5 mm; first question about any number: registered on what?

Questions

Thank you. Questions? · Three I would like your view on: · Do we apply 1.5 : 1 at the orbital floor, or fit by eye? · Which maxillary registration does the department use for treatment-change tracings? · Should the canal and third-molar bud stay on our 2D mandibular list?

Backup: the key tables

Structure	n	Mean	SD	Min	Max
Clinoid process	50	2.64	1.43	0.58	7.50
Anterior cranial base	50	2.52	1.13	0.53	4.90
Ethmoid triad	50	2.52	2.01	0.23	11.99
Posterior key ridge	50	0.97	0.43	0.47	3.12
Anterior key ridge	40	0.95	0.57	0.15	2.56
Lateral border of orbit	50	1.27	0.72	0.41	4.29
Orbital floor	47	1.40	0.98	0.35	4.39
Infraorbital foramen	34	1.67	1.31	0.52	5.79
PTM vertical	50	2.39	1.69	0.15	8.81
PTM horizontal	50	1.52	0.99	0.30	4.10
Palate, superior border	50	1.05	0.52	0.43	2.90

Backup: the key tables (continued)

Structure	n	Mean	SD	Min	Max
Palate, inferior border	50	1.14	0.60	0.31	3.34
Palate, internal	50	0.85	0.43	0.28	2.45
ANS vertical	50	1.29	0.86	0.15	4.82
ANS horizontal	50	1.43	1.14	0.15	5.09
A point	50	2.13	1.63	0.15	6.55
PNS	27	2.27	1.75	0.32	9.10
Landmark	Best fit X / Y	Implant X / Y	Structural X / Y		
PNS	−1.18 / +0.36	−2.34 / −0.70	−1.81 / −1.02		
ANS (Harvold)	+0.41 / +0.37	−0.91 / −1.56	−0.34 / −1.62		
ANS	+0.62 / +0.28	−0.63 / −1.62	−0.12 / −1.72		
A point	+0.14 / −0.47	−0.78 / −2.33	−0.65 / −2.43		

Backup: the key tables (continued)

Structure	n	Mean	SD	Min	Max
U6 cusp	+3.02 / −3.40	+1.31 / −4.75	+2.06 / −4.99		
U6 apex	+0.92 / −2.33	−0.42 / −3.80	+0.22 / −4.02		
UI edge	+1.22 / −1.40	−0.46 / −3.39	+0.15 / −3.39		
UI apex	−0.08 / −2.60	−1.17 / −4.36	−0.86 / −4.48		

- Doppel Table II, maximum deviation after implant superimposition, mm:
- Doppel Table III: palatal plane T1–T2 +0.92° ± 2.10° (printed Minimum 8°, Maximum −5°, swapped); SN line −0.56° ± 2.19° (printed 9°, −6°). Positive = downward and backward (Results text).